Machine Learning Challenge for "Merit Prize 2025–26"

Portuguese Fake News Detection

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Abstract

This work addresses the Portuguese Fake News Detection Challenge using the FakeNews-PT dataset, focusing on binary classification of news articles via machine learning models with TF-IDF features and interpretability techniques.

1 Model Training & Evaluation

a) Feature Extraction

We extracted TF-IDF features from the text using `TfidfVectorizer(max_features=5000, smooth_idf=True, min_df=10, max_df=0.9, ngram_range=(1,3), stop_words=portuguese_stopwords)`.

This configuration applied IDF smoothing for out-of-vocabulary terms, filtered rare terms (min_df=10) and overly common terms (max_df=0.9), and limited the vocabulary to the top 5000 features while capturing unigrams through trigrams. Portuguese stopwords were removed to reduce noise, and n-grams enabled detection of multi-word phrases like "fake news" that carry semantic meaning beyond single tokens.

b) Hyperparameter Tuning

We performed a grid search with 5-fold cross-validation for each classifier to tune key hyperparameters. For Logistic Regression with L2 regularization, we varied the inverse regularization strength $C \in \{0.01, 0.1, 1, 10\}$ and the class weight in {None, balanced}; the best configuration was $C = 10$ with no class weighting, achieving a mean validation accuracy of approximately 0.90. For L1-regularized Logistic Regression, we used the same grid over C and class weights, with the best setting $C = 1$ and balanced class weights.

For the MLP, we tuned the hidden layer sizes {(50), (100), (50,25), (100,50)}, the L_2 penalty $\alpha \in \{10^{-4}, 10^{-3}\}$, the learning rate $learning_rate_init = 10^{-3}$, and the number of SVD components {100, 200}; the best model used ReLU activation, hidden layers $(100,)$, $\alpha = 10^{-4}$, and 200 SVD components. For the Decision Tree, we varied the criterion in {gini, entropy}, the maximum depth in {None, 10, 20}, and the minimum samples per leaf in {1, 2, 5}; the best tree used the gini criterion with no depth limit and min_samples_leaf=1.

For Gaussian Naive Bayes, we tuned the variance smoothing parameter in $\{10^{-9}, 10^{-8}\}$ and the number of SVD components in $\{100, 200\}$; the best configuration used $var_smoothing = 10^{-9}$ and 200 components.

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c) Comparison Table and ROC Curve

The best configuration for each model was then evaluated on the test set, and their metrics are summarized below in Table 1.

Table 1: Table 1: Test set performance of best model configurations.

Model	Accuracy	Precision	Recall	F1
Logistic Regression (L2)	0.9002	0.9003	0.9002	0.9002
Logistic Regression (L1)	0.8959	0.8960	0.8959	0.8959
MLP	0.8957	0.8957	0.8957	0.8957
Decision Tree	0.8288	0.8290	0.8288	0.8287
Gaussian NB	0.7346	0.7400	0.7344	0.7330

The table shows that the Logistic Regression had the best performance in the test set with L2 performing slightly better. So we plotted the ROC and calculated the area under said curve for that model which can be seen below in Figure 1.

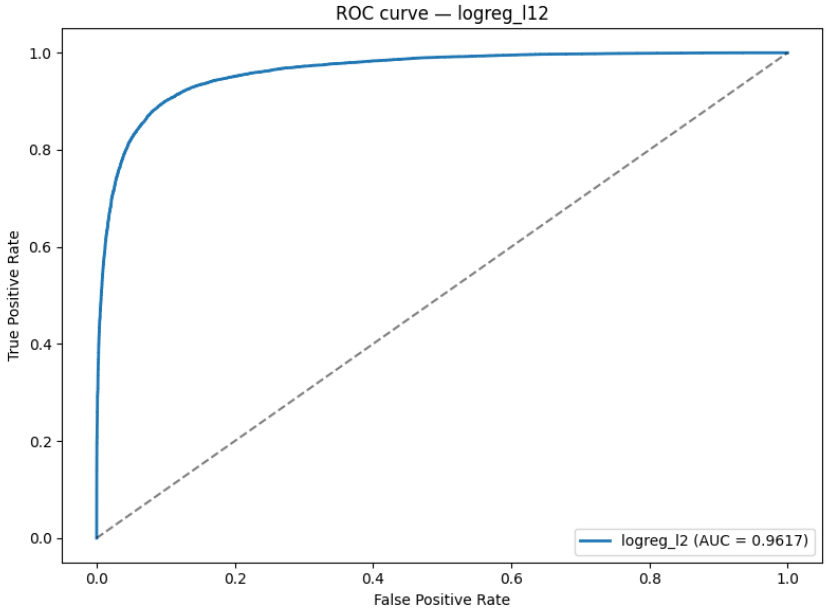


Figure 1: Fig 1 :ROC curve for best Logistic Regression (L2) model.

2 Model Interpretation

a) Top 10 words for fake and real news for the best Logistic Regression model

As we can see in Figure 2, the best Logistic Regression model is with the L2 regularization. We extracted the features that are the biggest representative of real or fake news. In the Figure 2, its shown the bar plot of said features.

b) Comparing L1 and L2 Regularization

b.1 How many features have non-zero weights in each?

Logistic Regression with L1 regularization has 2155 non-zero features. We can observe that L1 regularization (Lasso) produces sparse models: many coefficients are exactly zero.

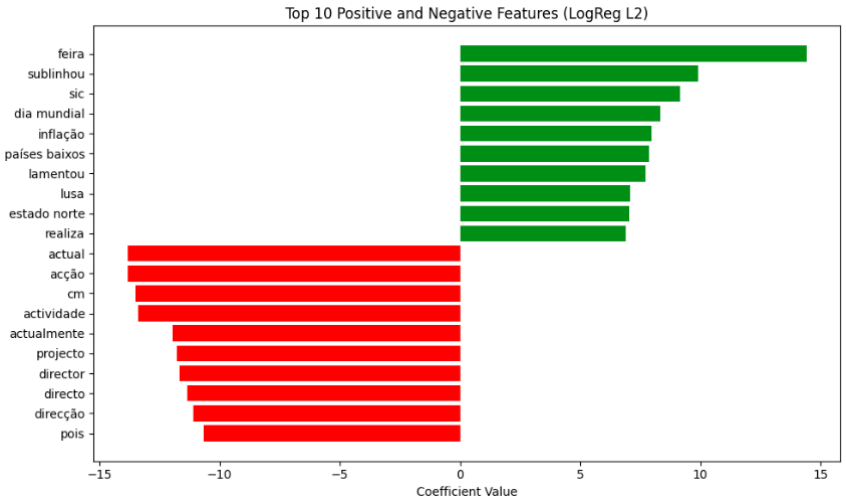


Figure 2: Top 10 Positive and Negative Features for the Logistic Regression with L2 regularization

Logistic Regression with L2 regularization has 5000 non-zero features (all of the features are non-zero). We can observe that L2 regularization (Ridge) produces dense models: most coefficients are non-zero (but their values may be small).

b.2 What does this tell you about feature selection?

L1 regularization selects features by setting many weights to zero, automatically choosing a subset of important words. L2 regularization shrinks all weights but does not remove features.

b.3 When would you prefer L1 vs L2 for text classification?

L1: When you want automatic feature selection, a simpler model, or interpretability. Its mostly used when only some terms really matters

L2: When you expect many features to contribute a little, or want to avoid overfitting without removing features.

In conclusion: We prefer L1 regularization for high-dimensional, sparse data where only some features matter and the features aren't strongly correlated. We prefer L2 regularization for high-dimensional data where all features may matter. It's useful when we can't disregard many features like, for example, to classify the authenticity of news.

c) Explanations with LIME for the Logistic Regression model

To produce the explanations with LIME, we followed the tutorial provided in the statement and obtained the results shown in Table 2.

Table 2: LIME explanation (Logistic Regression L2)

	Probabiblity		
Validation Set ID	Fake	Real	Real Label
2921	0.00114812	0.99885188	Real
2437	0.9988932	0.0011068	Fake
5557	0.00452681	0.99547319	Real
1697	0.84205924	0.15794076	Fake

d) Explanations with LIME for the MLP

We repeated the experiment but using the Multilayer Perceptron and got the results shown in Table 3.

Table 3: Table 3: LIME explanation (Multilayer Perceptron)

	Probabiblity		
Validation Set ID	Fake	Real	Label
2921	0.00528068	0.99471932	Real
2437	0.86318719	0.13681281	Fake
5557	0	1	Real
1697	0.999725311	0.000274689488	Fake

Then we selected 1000 random samples and got an accuracy of 87.9% . Looking at Table 3, one would assume that the accuracy of our model would be near perfect, since it classified all the samples correctly and with great certainty.

In any case, the result produced by the random samples does not greatly contradict what was expected, since the accuracy is reasonably high.

3 Clustering

a) & b) K-Means and cluster inspection

We applied K-Means with K = 5 to the training set , resulting in five clusters whose top terms show thematic correlation, as shown in Table 4. To assign semantic labels to each cluster, we used zero-shot classification model using arbitrary labels that we previously defined. Because the K-Means algorithm may converge to different centroids, we can get slightly different results across runs. Consequently, the zero-shot classification model will also assign different labels to the clusters.

Note: cluster 1 assigned label is 'entertainment news'. We assume that zero-shot is considering world wide known names like 'putin' and 'wagner' to be "celebrity" names and the strong words used in the articles are to "catch peoples attention" just like entertainment news do.

a.1 Cluster 0

Top 10 terms: ministro, costa, governo, ps, antónio, antónio costa, psd, primeiro ministro, primeiro, partido.

3 Closest documents:
→ [1] Doc 39852 (REAL) - Distance: 0.8649 - Predicted topic: political news (confidence: 0.30)
Preview: maioria conselheiros defende remodelação presidente marcelo anunciou convocação conselho estado final altura chamará belém partidos maioria conselheir...

→ [2] Doc 31138 (REAL) - Distance: 0.8679 - Predicted topic: political news (confidence: 0.21)
Preview: ana gomes governo fabricar populismos militante base gosta sublinhar ana gomes vozes independentes ps vê bons olhos braço ferro marcelo cala críticas ...

→ [3] Doc 2752 (REAL) - Distance: 0.8744 - Predicted topic: political news (confidence: 0.22)
Preview: costa marcelo reuniram belém antes comunicação pr país primeiro ministro reuniu palácio belém presidente república seguida comunicou país posição sobr...

→ ASSIGNED LABEL: 'political news (real)'
→ Topic votes: ['political news', 'political news', 'political news']
→ Cluster size: 7804 documents (3451 fake, 4353 real)

a.2 Cluster 1

Top 10 terms: ucrânia, rússia, russo, guerra, putin, wagner, moscovo, russa, kiev, ucraniano.

3 Closest documents:
→[1] Doc 22310 (REAL) - Distance to centroid: 0.8132 - Predicted topic: entertainment news (confidence: 0.18)
Preview: líder grupo wagner quer batalha final clarificar situação ucrânia "em teoria rússia pôs fim isto" yevgeny prigojin quer rússia declare cumpridos objet...

→[2] Doc 7742 (REAL) - Distance to centroid: 0.8175 - Predicted topic: entertainment news (confidence: 0.18)
Preview: rússia adormeceu acordou alvoroço rebelião grupo paramilitar wagner controla rostov moscovo alvo prigozhin promete ir fim garante homens renderão puti...

→[3] Doc 15729 (REAL) - Distance to centroid: 0.8181 - Predicted topic: political news (confidence: 0.19)
Preview: compreender conflito podemos esperar contraofensiva ucraniana contraofensiva ucraniana dimensão mau sinal demorar tanto pode fazer parte estratégia uc...

→ ASSIGNED LABEL: 'entertainment news (real)'
→ Topic votes: ['entertainment news', 'entertainment news', 'political news']
→ Cluster size: 2187 documents (125 fake, 2062 real)

a.3 Cluster 2

Top 10 terms: portugal, pessoas, saúde, anos, feira, ainda, dia, pode, lisboa, ter.

3 Closest documents:
→[1] Doc 2215 (FAKE) - Distance to centroid: 0.1471 - Predicted topic: health news (confidence: 0.18)
Preview: morrem vacinados vacinados...

→[2] Doc 32791 (FAKE) - Distance to centroid: 0.1471 - Predicted topic: entertainment news (confidence: 0.20)
Preview: aracely henriquez raptada agredida brutalmente george floyd"...

→[3] Doc 39521 (FAKE) - Distance to centroid: 0.1471 - Predicted topic: economic news (confidence: 0.16)
Preview: "relógio...

→ ASSIGNED LABEL: 'economic news (real)'
→ Topic votes: ['health news', 'entertainment news', 'economic news']
→ Cluster size: 29135 documents (13514 fake, 15621 real)

a.4 Cluster 3

Top 10 terms: anos, mãe, vida, jovem, mulher, homem, crianças, pessoas, ter, filho.

3 Closest documents:
→[1] Doc 48388 (FAKE) - Distance to centroid: 0.9092 - Predicted topic: entertainment news (confidence: 0.34)
Preview: casos curiosos crianças afirmam lembrar vidas passadas acontece morremos vamos céu inferno viramos comidinha verme voltamos vida outro corpo ciência r...

→[2] Doc 32141 (FAKE) - Distance to centroid: 0.9092 - Predicted topic: entertainment news (confidence: 0.34)
Preview: casos curiosos crianças afirmam lembrar vidas passadas acontece morremos vamos céu inferno viramos comidinha verme voltamos vida outro corpo ciência r...

→[3] Doc 2718 (FAKE) - Distance to centroid: 0.9092 - Predicted topic: entertainment news (confidence: 0.34)
Preview: casos curiosos crianças afirmam lembrar vidas passadas acontece morremos vamos céu inferno viramos comidinha verme voltamos vida outro corpo ciência r...

→ ASSIGNED LABEL: 'entertainment news (fake)'
→ Topic votes: ['entertainment news', 'entertainment news', 'entertainment news']

→ Cluster size: 6995 documents (5617 fake, 1378 real)

a.5 Cluster 4

Top 10 terms: euros, milhões, milhões euros, mil, mil milhões, mil euros, ano, valor, banco, portugal

3 Closest documents:
→[1] Doc 7145 (FAKE) - Distance to centroid: 0.8157 - Predicted topic: political news (confidence: 0.17)
Preview: milhões euros fogem offshores...

→[2] Doc 40477 (REAL) - Distance to centroid: 0.8177 - Predicted topic: economic news (confidence: 0.17)
Preview: infraestruturas portugal metropolitano lisboa maiores beneficiários prr total plano recuperação resiliência contabilizam beneficiários projetos metrop...

→[3] Doc 4527 (REAL) - Distance to centroid: 0.8189 - Predicted topic: economic news (confidence: 0.28)
Preview: economia portuguesa passa défice excedente externo milhões regulador afirma aumento exportações reflete sobretudo contributo viagens turismo cujas exp...

→ ASSIGNED LABEL: 'economic news (fake)'
→ Topic votes: ['political news', 'economic news', 'economic news']
→ Cluster size: 4428 documents (2497 fake, 1931 real)

a.6 Overall Cluster inspection

Each cluster exhibits internally coherent thematic patterns, indicating that the feature space successfully captured meaningful structure within the dataset.

However, when examining the distribution of class labels (real versus fake news) within each cluster, only two clusters demonstrate a clear majority of either real or fake news articles. The remaining three clusters display an approximately balanced (near 50/50) distribution of labels.

This outcome suggests that while the clustering procedure effectively groups articles based on shared linguistic or semantic characteristics, these characteristics do not consistently align with the ground-truth classification of news authenticity.

b) Visualize clusters in 2D using PCA.

We standardized the data before performing PCA, so we could detect the real variance of the features. As we can observe on Figure 3, we can see clearly that there are 3 clusters formed: at down left a very dense cluster , at down right a sparse cluster and at top left a very small cluster.

With the exception of the top left cluster, the data set seems aligned with the PCA Component 1, suggesting the the PCA Component 2 does not impact as much as PCA Component 1 in describing the data set.

Effectively we cannot really detect a separation of clusters after applying the PCA. Only cluster 4 seems to be grouped in the on the the down right cluster.

Assessing the data set by the true label, only the cluster at down left has both real and fake labels. Meaning we need more components to explain the data in a satisfactory way.

It is important to mention that these two components only explain 16.5% of the whole data set. This indicates that, indeed, we need more components to explain the data and that the detection of real and fake news is not a simple task but rather a complex one.

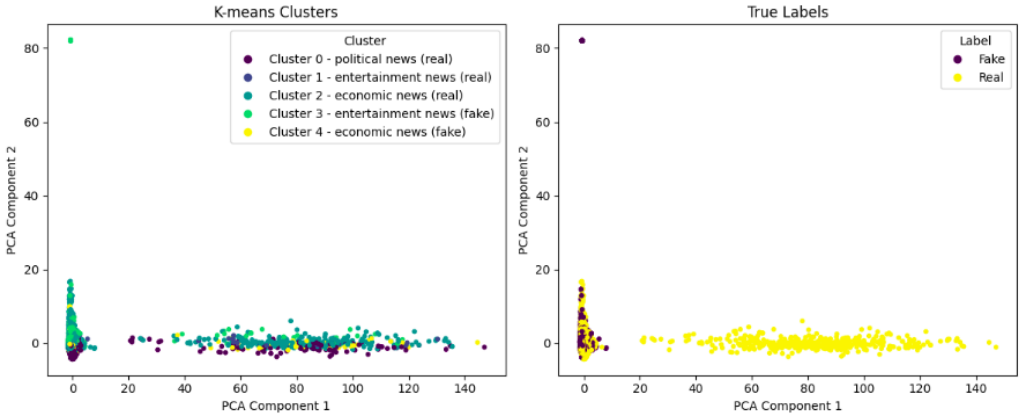


Figure 3: PCA visualization by cluster assignment and by true label

References

[1] Python Tutorials for Digital Humanities. (2021) TF-IDF in Python with Scikit Learn (Topic Modeling for DH 02.03). YouTube. video-URL

[2] A. Wichert, L. Sa-Couto. Machine Learning - A Journey to Deep Learning with Exercises and Answers

[3] Mitchell, T.M. (1997) Machine Learning. New York, NY: McGraw-Hill.